EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Gravitational Force

1. If the distance between two masses is doubled, the gravitational attraction between them
(A) Is doubled (B) Becomes four times
(C) Is reduced to half (D) Is reduced to a quarter

PGT033

2. The earth (mass = 6×10^{24} *kg*) revolves round the sun with angular velocity 2×10^{-7} *rad/s* in a circular orbit of radius 1.5×10^{8} *km*. The force exerted by the sun on the earth, in newtons is :-
(A) 18×10^{25} (B) zero (C) 27×10^{39} (D) 36×10^{21}

PGT034

3. The distance of the centres of moon and earth is *D*. The mass of earth is 81 times the mass of the moon. At what distance from the centre of the earth, the gravitational force will be zero
(A) $\frac{D}{2}$ (B) $\frac{2D}{3}$ (C) $\frac{4D}{3}$ (D) $\frac{9D}{10}$

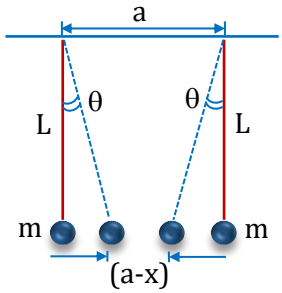
PGT035

4. Four particles of equal mass *M* move along a circle of radius *R* under the action of their mutual gravitational attraction. The speed of each particle is -
(A) $\frac{GM}{R}$ (B) $\sqrt{\left[2\sqrt{2}\frac{GM}{R}\right]}$ (C) $\sqrt{\left[\frac{GM}{R}\left(2\sqrt{2}+1\right)\right]}$ (D) $\sqrt{\left[\frac{GM}{R}\left(\frac{2\sqrt{2}+1}{4}\right)\right]}$

PGT038

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5. Two small balls of mass *m* each are suspended side by side by two equal threads of length *L* as shown in the figure. If the distance between the upper ends of the threads be *a*, the angle θ that the threads will make with the vertical due to attraction between the balls is

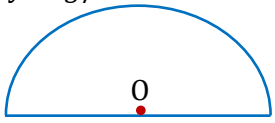


(A) $\tan^{-1}\frac{(a-x)g}{mG}$ (B) $\tan^{-1}\frac{mG}{(a-x)^2 g}$ (C) $\tan^{-1}\frac{(a-x)^2 g}{mG}$ (D) $\tan^{-1}\frac{(a^2-x^2)g}{mG}$

PGT039

Gravitation Intensity, Potential and Potential Energy

6. Find the field strength at the centre of a thin semicircular ring of radius *R* and mass *M* as shown in figure. Take the linear mass density λ *kg/m* -

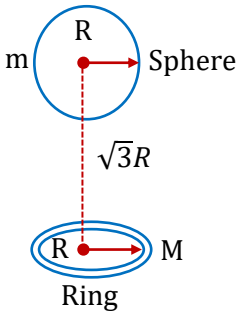


(A) $\frac{G\lambda}{R}$ (B) $\frac{4G\lambda}{R}$ (C) $\frac{2G\lambda}{R}$ (D) $\frac{G\lambda}{2R}$

PGT036

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7. A uniform ring of mass M is lying at a distance $\sqrt{3}R$ from the centre of a uniform sphere of mass m just below the sphere as shown in the figure where R is the radius of the ring as well as that of the sphere. Then gravitational force exerted by the ring on the sphere is:



(A) $\frac{GMm}{8R^2}$ (B) $\frac{GMm}{3R^2}$

(C) $\sqrt{3}\frac{GMm}{R^2}$ (D) $\sqrt{3}\frac{GMm}{8R^2}$

PGT044

8. A spherical cavity of radius $\frac{R}{2}$ is made in a sphere of radius R and mass M. The centre of the cavity is at a distance $\frac{R}{2}$ from the centre of the sphere. The intensity of gravitational field at the centre of cavity is :-

(A) 0 (B) $\frac{GM}{R^2}$ (C) $\frac{GM}{2R^2}$ (D) $\frac{3GM}{2R^2}$

PGT040

9. Let E_1 and E_2 denotes gravitational field at distance 'r_1' and 'r_2' from axis of infinitely long solid cylinder of radius 'R'. Which of the following must hold true.

(A) $E_1 < E_2$ if $r_1 < r_2 < R$

(B) $E_1 > E_2$ if $R < r_1 < r_2$

(C) $E_1 > E_2$ if $r_1 = R - E$, $r_2 = R + E$ (E is positive constant $< R$)

(D) All of the above

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PGT041

10. If the radius of the earth be increased by a factor of 5, by what factor its density be changed to keep the value of g the same?

(A) 1/25 (B) 1/5 (C) $1/\sqrt{5}$ (D) 5

PGT037

11. The masses and radii of the earth and moon are M_1, R_1 and M_2, R_2 respectively. Their centres are distance d apart. The minimum velocity with which a particle of mass m should be projected from a point midway between their centres so that it escapes to infinity is

(A) $2\sqrt{\frac{G}{d}(M_1+M_2)}$ (B) $2\sqrt{\frac{2G}{d}(M_1+M_2)}$ (C) $2\sqrt{\frac{Gm}{d}(M_1+M_2)}$ (D) $2\sqrt{\frac{Gm(M_1+M_2)}{d(R_1+R_2)}}$

PGT043

12. A particle is fired vertically from the surface of the earth with a velocity kv_e, where v_e is the escape velocity and $k < 1$. Neglecting air resistance and assuming earth's radius as R_e. Calculate the height to which it will rise from the surface of the earth.

(A) $\left(\frac{R_e k^2}{1-k^2}\right)$ (B) $\left(\frac{R_e k}{1-k^2}\right)$ (C) $\left(\frac{R_e k^2}{1-k}\right)$ (D) None

PGT048

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Kepler's Law and Satellite Motion

13. The escape velocity from earth is V_{es}. A body is projected with velocity $2V_{es}$ with what constant velocity will it move in the inter planetary space

(A) v_{es} (B) $3v_{es}$ (C) $\sqrt{3}v_{es}$ (D) $\sqrt{5}v_{es}$

PGT046

14. V_e and V_p denotes the escape velocity from the earth and another planet having twice the radius and the same mean density as the earth. Then

(A) $v_e = v_p$ (B) $v_e = v_p/2$ (C) $v_e = 2v_p$ (D) $v_e = v_p/4$

PGT047

15. A satellite can be in a geostationary orbit around earth at a distance r from the centre. If the angular velocity of earth about its axis doubles, a satellite can now be in a geostationary orbit around earth if its distance from the centre is :-

(A) $\frac{r}{2}$ (B) $\frac{r}{2\sqrt{2}}$ (C) $\frac{r}{(4)^{1/3}}$ (D) $\frac{r}{(2)^{1/3}}$

PGT054

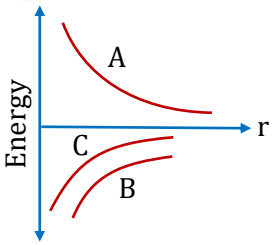
16. A satellite of mass $5M$ orbits the earth in a circular orbit. At one point in its orbit, the satellite explodes into two pieces, one of mass M and the other of mass $4M$. After the explosion the mass M ends up travelling in the same circular orbit, but in opposite direction. After explosion the mass $4M$ is

(A) In a circular orbit
(B) unbound
(C) elliptical orbit
(D) data is insufficient to determine the nature of the orbit.

PGT055

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17. The figure shows the variation of energy with the orbit radius of a body in circular planetary motion. Find the correct statement about the curves A, B and C



(A) A shows the kinetic energy, B the total energy and C the potential energy of the system.
(B) C shows the total energy, B the kinetic energy and A the potential energy of the system.
(C) C and A are kinetic and potential energies respectively and B is the total energy of the system.
(D) A and B are kinetic and potential energies and C is the total energy of the system.

PGT050

18. A satellite of mass m, initially at rest on the earth, is launched into a circular orbit at a height equal to the radius of the earth. The minimum energy required is

(A) $\frac{\sqrt{3}}{4}mgR$ (B) $\frac{1}{2}mgR$ (C) $\frac{1}{4}mgR$ (D) $\frac{3}{4}mgR$

PGT051

19. A space ship of mass m is in circular orbit of radius $2R_e$ about the earth of mass M and radius R_e. Energy required to transfer the space ship to circular orbit of radius $3R_e$ is :

(A) $\frac{GMm}{8R_e}$ (B) $\frac{GMm}{4R_e}$ (C) $\frac{GMm}{24R_e}$ (D) $\frac{GMm}{12R_e}$

PGT052

For More Join: @IITJEE_Advanced

20. An earth satellite is moved from one stable circular orbit to another larger and stable circular orbit. The following quantities increase for the satellite as a result of this change:-
(A) gravitational potential energy (B) angular velocity
(C) linear orbital velocity (D) centripetal acceleration

PGT053

21. Two identical satellites are at R and $7R$ away from earth surface, the wrong statement is (R = Radius of earth)
(A) Ratio of total energy will be 4
(B) Ratio of kinetic energies will be 4
(C) Ratio of potential energies will be 4
(D) Ratio of total energy will be 4 but ratio of potential and kinetic energies will be 2

PGT057

22. A planet of mass m is in an elliptical orbit about the sun ($m << M_{sun}$) with an orbital period T. If A be the area of orbit, then its angular momentum would be :
(A) $\frac{2mA}{T}$ (B) mAT (C) $\frac{mA}{2T}$ (D) $2mAT$

PGT063

23. A planet revolves about the sun in elliptical orbit. The areal velocity $\left(\frac{dA}{dt}\right)$ of the planet is $4.0 \times 10^{16}\ m^2/s$. The least distance between planet and the sun is $2 \times 10^{12}\ m$. Then the maximum speed of the planet in km/s is :
(A) 10 (B) 20 (C) 40 (D) None of these

PGT066

24. Satellites A and B are orbiting around the earth in orbits of ratio R and $4R$ respectively. The ratio of their areal velocities is :
(A) 1 : 2 (B) 1 : 4 (C) 1 : 8 (D) 1 : 16

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PGT067

25. The time period of a satellite of earth is 5 hours. If the separation between the centre of earth and the satellite is increased to 4 times the previous value, the new period will become-
(A) 10 h (B) 80 h (C) 40 h (D) 20 h

PGT070

26. The Sun travels in approximately circular orbit of radius R around the center of the galaxy and completes one revolution in time T. The Earth also revolves around the Sun in time t. Assume orbit of the Earth to be a circle of radius r ($r << R$) and whole mass of the galaxy centered on its center. By using only these given information's, find an expression for the ratio of the mass of the galaxy to that of the Sun.
(A) $\left(\frac{R}{r}\right)^3\left(\frac{t}{T}\right)^2$ (B) $\left(\frac{R}{r}\right)^3\left(\frac{T}{t}\right)^2$ (C) $\left(\frac{R}{r}\right)^2\left(\frac{t}{T}\right)^3$ (D) $\left(\frac{R}{r}\right)^2\left(\frac{T}{t}\right)^3$

PGT071

27. In the following four periods
(i) Time of revolution of a satellite just above the earth's surface (T_{st})
(ii) Period of oscillation of mass inside the tunnel bored along the diameter of the earth (T_{ma})
(iii) Period of simple pendulum having a length equal to the earth's radius in a uniform field of $9.8\ N/kg$ (T_{sp})
(iv) Period of an infinite length simple pendulum in the earth's real gravitational field (T_{is})
(A) $T_{st} > T_{ma}$ (B) $T_{ma} > T_{st}$ (C) $T_{sp} < T_{is}$ (D) $T_{st} = T_{ma} = T_{sp} = T_{is}$

PGT072

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Two tunnels are dug from one side of the Earth's surface to the other side, one along a diameter and the other along a chord. Now two particle are dropped from one end of each of the tunnels. Both the particles oscillate simple harmonically along the tunnels. Let T_1 and T_2 be the time period, v_1 and v_2 be the maximum speed of the particle in the two tunnels. Then :-

(A) $T_1 = T_2$ (B) $T_1 > T_2$ (C) $v_1 = v_2$ (D) $v_1 > v_2$

PGT042

2. If a satellite orbits as close to the earth's surface as possible,

(A) its speed is maximum

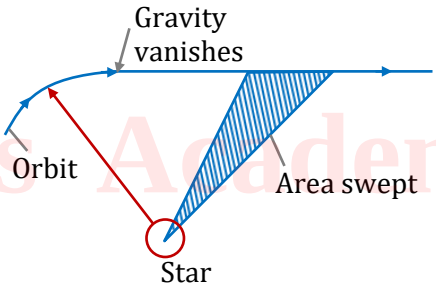
(B) time period of its rotation is minimum

(C) the total energy of the 'earth plus satellite' system is minimum

(D) the total energy of the 'earth plus satellite' system is maximum

PGT049

3. A planet is orbiting a star when for no apparent reason the star's gravity suddenly vanishes. After which planet moves in a straight line. Mark the **CORRECT** statement(s) :



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(A) Newton's first law is obeyed on planet after gravity vanishes

(B) Kepler's law of areas is obeyed only till the planet is in gravity of star

(C) Kepler's law of areas is obeyed even after gravity vanishes

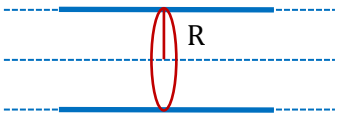
(D) Angular momentum of planet about centre of star is conserved through out its motion

PGT065

COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NO. 4 TO 6

Consider a hypothetical planet which is very long and cylindrical. The density of the planet is ρ and its radius is R.



4. What is the possible orbital speed of the satellite in moving around the planet in circular orbit in a plane which is perpendicular to the axis of planet?

(A) $R\sqrt{\pi G\rho}$ (B) $2R\sqrt{\pi G\rho}$ (C) $R\sqrt{2\pi G\rho}$ (D) $R\sqrt{\frac{G\rho}{2\pi}}$

PGT059

For More Join: @IITJEE_Advanced

5. If an object is projected radially outwards from the surface such that it reaches upto a maximum distance of $3R$ from the axis then what should be the speed of projection?

(A) $R\sqrt{\frac{2}{3}\pi\rho G}$ (B) $2R\sqrt{\pi\rho G\ell n3}$ (C) $R\sqrt{\frac{4}{3}\pi\rho G}$ (D) $R\sqrt{\frac{2}{3}\pi\rho G\ell n3}$

PGT060

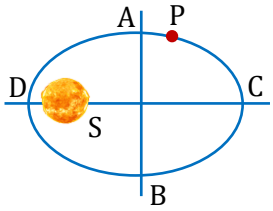
6. Assume that the planet is rotating about its axis with time period T. How far from the axis of the planet do the **geo synchronous telecommunication satellites** orbit are?

(A) $RT\sqrt{\pi G\rho}$ (B) $2RT\sqrt{\pi G\rho}$ (C) $RT\sqrt{2\pi G\rho}$ (D) $RT\sqrt{\frac{G\rho}{2\pi}}$

PGT061

PARAGRAPH FOR QUESTION NO. 7 AND 8

Figure shows the orbit of a planet P around the sun S. AB and CD are the minor and major axes of the ellipse.



7. If t_1 is the time taken by the planet to travel along ACB and t_2 the time along BDA, then

(A) $t_1 = t_2$ (B) $t_1 > t_2$

(C) $t_1 < t_2$ (D) nothing can be concluded

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PGT068

8. If U is the potential energy and K is the kinetic energy then $|U| > |K|$ at

(A) Only D (B) Only C (C) Both D and C (D) Neither D nor C

PGT069

MATRIX MATCH TYPE QUESTION

9. In elliptical orbit of a planet, as the planet moves from apogee position to perigee position,

	Column-I		**Column-II**
(A)	Speed of planet	(P)	Remains same
(B)	Distance of planet from centre of Sun	(Q)	Decreases
(C)	Potential energy	(R)	Increases
(D)	Angular momentum about centre of Sun	(S)	Can not say

PGT064

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EXERCISE (S)

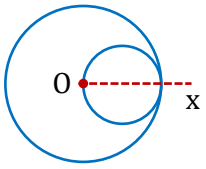
Gravitational Force

1. Two small balls of the same size and of mass m_1 and m_2 ($m_1 > m_2$) are tied by a thin light thread and dropped from a large height. Determine the tension T of the thread during the flight after the motion of the balls has become steady. (Air resistance force is same on two balls)
[Take $m_1 = 2kg$. $m_2 = 1\ kg, g = 10\ m/s^2$]

PGT076

Gravitation Intensity, Potential and Potential Energy

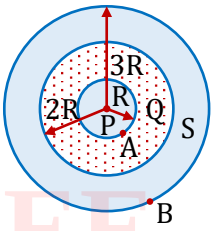
2. A sphere of radius R has its centre at the origin. It has a uniform mass density r_o except that there is a spherical hole of radius $r = R/2$ whose centre is at $x = R/2$ as in fig.



(a) Find gravitational field at points on the axis for $|x| > R$.
(b) Show that the gravitational field inside the hole is uniform, find its magnitude and direction.

PGT045

3. A solid sphere P and two hollow sphere Q and S are concentrically placed as shown in the figure. The radius of P. Q and S are R, $2R$ and $3R$ respectively. The density of P, Q and S are ρ_1, ρ_2 and ρ_3 respectively. It is given that the density of middle sphere is the arithmetic mean of the densities of other two spheres.



If the gravitational fields at points A and B are equal then find the ratio of $\frac{\rho_3}{\rho_1}$ and fill the value of $10\frac{\rho_3}{\rho_1}$.

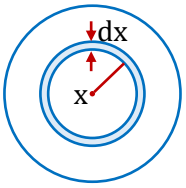
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PGT077

4. A hypothetical spherical planet of radius R and its density varies as $\rho = Kr$, where K is constant and r is the distance from the center. The pressure caused by gravitational pull inside ($r < R$) the planet at a distance r measured from its center is $\frac{K^2}{n^2} G\pi(R^4 - r^4)$. Find n^2.

PGT078

5. The density of a solid spherical planet of radius R is given as $\rho = \rho_0 r$, where ρ_0 = constant and r is distance measured from centre of planet. The acceleration due to gravity of this planet is half of maximum value at distance x from centre and also at a distance y from the centre. The value of $x + y$ is $R\sqrt{\frac{\alpha}{\beta}}$. Here α and β are single digit integer. Find the value of α-β?



(Where the fraction is in lowest form)

PGT079

Kepler's Law and Satellite Motion

6. A satellite close to the earth is in orbit above the equator with a period of rotation of 1.5 hours. If it is above a point P on the equator at some time, it will be above P again after time_____.

PGT058

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7. A satellite is moving in a circular orbit around the earth. The total energy of the satellite is $E = -2 \times 10^5 J$. The amount of energy to be imparted to the satellite to transfer it to a circular orbit where its potential energy is $U = -2 \times 10^5 J$ is equal to _____.

PGT056

8. A pair of stars rotates about a common center of mass. One of the stars has a mass M which is twice as large as the mass m of the other. Their centres are at a distance d apart, d being large compared to the size of either star.

(a) Derive an expression for the period of rotation of the stars about their common centre of mass in terms of d, m, G.

(b) Compare the angular momentum of the two stars about their common centre of mass by calculating the ratio L_m / L_M.

(c) Compare the kinetic energies of the two stars by calculating the ratio K_m/K_M.

PGT062

9. A comet is in elliptical orbit around the sun. In this orbit the comet's smallest distance from the sun is 72×10^6 m and its largest distance from the sun is 144×10^6 m. The ratio of comet's maximum speed to the minimum speed in the orbit is : (Neglect the presence of all bodies other than the sun and comet).

PGT080

10. A satellite is launched into a circular orbit of radius R around earth, while a second satellite is launched into a circular orbit of radius 1.02 R. The percentage difference in the time periods of the two satellites is

PGT126

11. A binary star consists of two stars A (mass 2.2 M_S) and B (mass 11 M_S), where M_S is the mass of the sun. They are separated by distance d and are rotating about their centre of mass, which is stationary. The ratio of the total angular momentum of the binary star to the angular momentum of star B about the centre of mass is

PGT082

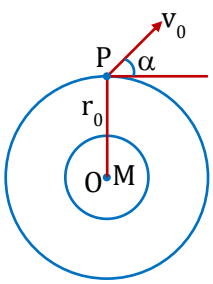
12. A satellite is launched in the equatorial plane in such a way that it can transmit signals upto 60° latitude on the earth. The orbital velocity of the satellite is found to be $\sqrt{\frac{GM}{\alpha R}}$. Find the value of α.

PGT083

13. A satellite of mass m is orbiting the Earth in a circular orbit of radius r. It starts losing energy due to small air resistance at the rate of $C\,J/\,s$. Then the time taken for the satellite to reach the Earth is $t = \frac{GMm}{nC}\left(\frac{1}{R_e} - \frac{1}{r}\right)$. Find n.

PGT084

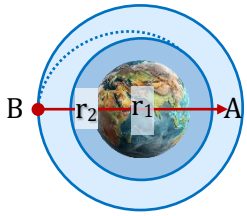
14. A satellite is projected into space from a point 'P' at a distance 'r_0' from the centre of the earth with an initial velocity $v_0 = \sqrt{\frac{GM}{r_0}}$ at an angle $\alpha = 37°$ with the horizontal at point 'P' as shown in the figure. It is found that the satellite is propelled into an elliptical orbit. If the eccentricity of the elliptical orbit is e, then value of $5e$ is ?



PGT085

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15. A spacecraft of mass m describes a circular orbit or radius r_1 around the earth.



(a) Find the additional energy ΔE which must be imparted to the spacecraft to transfer it to a circular orbit of larger radius r^2.

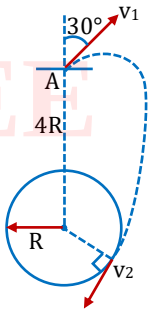
(b) If the transfer from one circular orbit to the other is executed by placing the spacecraft on a transitional semi elliptical path AB then find the amounts of energy ΔE_A and ΔE_B which must be imparted at A and B respectively.

PGT086

16. An astronaut in a circular orbit around earth observes a celestial body moving in a lower circular orbit around earth in same plane as his orbit and in the same sense. He observes that the body moves at a speed of 5 m/s relative to himself when it is closest to him. The minimum distance between him and the body if he is moving at a speed of 5000 m/s is α km. Fill the value of $\frac{\alpha}{4}$ in OMR sheet. (Mass of earth = 6×10^{24} kg and round off the value of α to the nearest integer).

PGT087

17. A particle is projected from point A, that is at a distance $4R$ from the centre of the earth, with speed v_1 in a direction making 30° with the line joining the centre of the earth and point A, as shown. Find v_1 if the particle passes grazing the surface of the earth. Consider gravitational interaction only between these two. Express your answer in the form $\frac{1000X}{\sqrt{2}} m/s$ and fill value of X.



(Use $\frac{GM}{R} = 6.4 \times 10^7\ m^2/s^2$)

PGT088

18. Gravitational acceleration on the surface of a planet is $\frac{g}{11}$, where g is the gravitational acceleration on the surface of the earth. The average mass density of the planet is $\frac{1}{4}$ times that of the Earth. If the escape speed on the surface of the earth is taken to be 11 kms^{-1}, the escape speed on the surface of the planet in kms^{-1} will be

PGT089

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EXERCISE - JEE (Main) PYQ

1. A rocket has to be launched from Earth in such a way that it never returns. If E is the minimum energy delivered by the rocket launcher, what should be the minimum energy that the launcher should have if the same rocket is to be launched from the surface of the moon ?
Assume that the density of the Earth and the moon are equal and that the Earth's volume is 64 times the volume of the moon : **[JEE (Main) 2019]**

(1) $\frac{E}{4}$ (2) $\frac{E}{16}$ (3) $\frac{E}{32}$ (4) $\frac{E}{64}$

PGT010

2. A satellite is moving in a low nearly circular orbit around the earth. Its radius is roughly equal to that of the Earth's radius R_e. By firing rockets attached to it, its speed is instantaneously increased in the direction of its motion so that is become $\sqrt{\frac{3}{2}}$ times larger. Due to this the farthest distance from the centre of the Earth that the satellite reaches is R, value of R is : **[JEE (Main) 2020]**

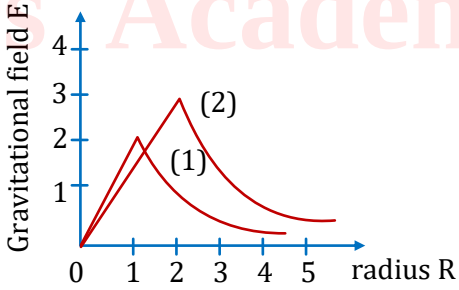
(1) $4R_e$ (2) $3R_e$ (3) $2R_e$ (4) $2.5R_e$

PGT011

3. Consider two solid spheres of radii $R_1 = 1m$, $R_2 = 2m$ and masses M_1 and M_2, respectively. The gravitational field due to sphere (1) and (2) are shown. The value of $\frac{M_1}{M_2}$ is :

[JEE (Main) 2020]

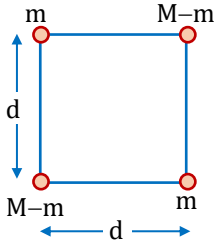
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(1) $\frac{1}{2}$ (2) $\frac{2}{3}$ (3) $\frac{1}{3}$ (4) $\frac{1}{6}$

PGT091

4. A body of mass $(2M)$ splits into four masses $\{m, M-m, m, M-m\}$, which are rearranged to form a square as shown in the figure. The ratio of $\frac{M}{m}$ for which, the gravitational potential energy of the system becomes maximum is $x : 1$. The value of x is ..... **[JEE (Main) 2021]**



PGT012

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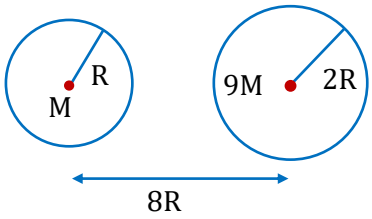
5. If R_E be the radius of Earth, then the ratio between the acceleration due to gravity at a depth 'r' below and a height 'r' above the earth surface is : (Given : $r < R_E$) **[JEE (Main) 2021]**

(1) $1-\frac{r}{R_E}-\frac{r^2}{R_E^2}-\frac{r^3}{R_E^3}$ (2) $1+\frac{r}{R_E}+\frac{r^2}{R_E^2}+\frac{r^3}{R_E^3}$ (3) $1+\frac{r}{R_E}-\frac{r^2}{R_E^2}+\frac{r^3}{R_E^3}$ (4) $1+\frac{r}{R_E}-\frac{r^2}{R_E^2}-\frac{r^3}{R_E^3}$

PGT092

6. Suppose two planets (spherical in shape) of radii R and $2R$, but mass M and $9M$ respectively have a centre to centre separation $8R$ as shown in the figure. A satellite of mass 'm' is projected from the surface of the planet of mass 'M' directly towards the centre of the second planet. The minimum speed 'v' required for the satellite to reach the surface of the second planet is $\sqrt{\frac{a}{7}\frac{GM}{R}}$ then the value of 'a' is _____.

[Given : The two planets are fixed in their position] **[JEE (Main) 2021]**



PGT093

7. Choose the incorrect statement from the following: **[JEE (Main) 2023]**

(1) The speed of satellite in a given circular orbit remains constant.

(2) For a planet revolving around the sun in an elliptical orbit, the total energy of the planet remains constant.

(3) When a body fall towards earth, the displacement of earth towards the body is negligible.

(4) The linear speed of a planet revolving around the sun remains constant.

PGT073

8. The weight of a body on the surface of the earth is 100 N. The gravitational force on it when taken at a height, from the surface of earth, equal to one-fourth the radius of the earth is:

[JEE (Main) 2023]

(1) 100 N (2) 64 N (3) 50 N (4) 25 N

PGT074

9. Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A: A pendulum clock when taken to Mount Everest becomes fast.

Reason R: The value of g (acceleration due to gravity) is less at Mount Everest than its value on the surface of earth.

In the light of the above statements, choose the most appropriate answer from the options given below **[JEE (Main) 2023]**

(1) Both **A** and **R** are correct but **R** is **NOT** the correct explanation of **A**

(2) Both **A** and **R** are correct and **R** is the correct explanation of **A**

(3) **A** is not correct but **R** is correct

(4) **A** is correct but **R** is not correct

PGT094

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10. Given below are two statements:

Statement I: Rotation of the earth shows effect on the value of acceleration due to gravity (g).

Statement II: The effect of rotation of the earth on the value of 'g' at the equator is minimum and that at the pole is maximum.

In the light of the above statements, choose the correct answer from the options given below.

[JEE (Main) 2023]

(1) Statement I is false but Statement II is true.

(2) Statement I is true but Statement II are false.

(3) Both Statement I and Statement II are true.

(4) Both Statement I and Statement II are false.

PGT095

11. Assume that the earth is a solid sphere of uniform density and a tunnel is dug along its diameter throughout the earth. It is found that when a particle is released in this tunnel, it executes a simple harmonic motion. The mass of the particle is 100 g. The time period of the motion of the particle will be (approximately)

(Take : $g = 10\ ms^{-2}$, radius of earth = 6400 km) **[JEE (Main) 2023]**

(1) 24 hours (2) 1 hour 24 minutes

(3) 1 hour 40 minutes (4) 12 hours

PGT096

12. A straight rod of length L extends from x = a to $x = L + a$. The gravitational force is exerts on a point mass 'm' at $x = 0$, if the mass per unit length of the rod is $A + Bx^2$, is given by:

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[JEE (Main) 2023]

(1) $Gm\left[A\left(\frac{1}{a+L}-\frac{1}{a}\right)-BL\right]$ (2) $Gm\left[A\left(\frac{1}{a}-\frac{1}{a+L}\right)+BL\right]$

(3) $Gm\left[A\left(\frac{1}{a+L}-\frac{1}{a}\right)+BL\right]$ (4) $Gm\left[A\left(\frac{1}{a}-\frac{1}{a+L}\right)-BL\right]$

PGT097

13. A space ship of mass $2 \times 10^4\ kg$ is launched into a circular orbit close to the earth surface. The additional velocity to be imparted to the space ship in the orbit to overcome the gravitational pull will be (if $g = 10\ m/s^2$ and radius of earth = 6400 km) **[JEE (Main) 2023]**

(1) $11.2\left(\sqrt{2}-1\right)km/s$ (2) $7.9\left(\sqrt{2}-1\right)km/s$

(3) $8\left(\sqrt{2}-1\right)km/s$ (4) $7.4\left(\sqrt{2}-1\right)km/s$

PGT098

14. A simple pendulum doing small oscillations at a place R height above earth surface has time period of $T_1 = 4\ s$. T_2 would be it's time period if it is brought to a point which is at a height $2R$ from earth surface. Choose the correct relation [R = radius of Earth]: **[JEE (Main) 2024]**

(1) $T_1 = T_2$ (2) $2T_1 = 3T_2$ (3) $3T_1 = 2T_2$ (4) $2T_1 = T_2$

PGT100

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15. Match **list-I** with **list-II**:

	List-I		**List-II**
(A)	Kinetic energy of planet	(I)	$-\frac{GMm}{a}$
(B)	Gravitation Potential energy of Sun-planet system.	(II)	$\frac{GMm}{2a}$
(C)	Total mechanical energy of planet	(III)	$\frac{Gm}{r}$
(D)	Escape energy at the surface of planet for unit mass object	(IV)	$-\frac{GMm}{2a}$

(Where a = radius of planet orbit, r = radius of planet, M = mass of Sun, m = mass of planet)
Choose the correct answer from the options given below: **[JEE (Main) 2024]**

(1) (A) – II, (B) – I, (C) – IV, (D) – III
(2) (A) – III, (B) – IV, (C) – I, (D) – II
(3) (A) – I, (B) – IV, (C) – II, (D) – III
(4) (A) – I, (B) – II, (C) – III, (D) – IV

PGT101

16. An astronaut takes a ball of mass m from earth to space. He throws the ball into a circular orbit about earth at an altitude of 318.5 km. From earth's surface to the orbit, the change in total mechanical energy of the ball is $x\frac{GM_em}{21R_e}$. The value of x is (take R_e = 6370 km): **[JEE (Main) 2024]**

(1) 11 (2) 9 (3) 12 (4) 10

PGT102

17. A satellite of 10^3 kg mass is revolving in circular orbit of radius $2R$. If $\frac{10^4 R}{6}$ J energy is supplied to the satellite, it would revolve in a new circular orbit of radius :
(use $g = 10 m/s^2$, R = radius of earth) **[JEE (Main) 2024]**

(1) 2.5 R (2) 3 R (3) 4 R (4) 6 R

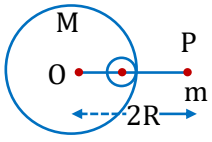
PGT103

18. A light planet is revolving around a massive star in a circular orbit of radius R with a period of revolution T. If the force of attraction between planet and star is proportional to $R^{-3/2}$ then choose the correct option : **[JEE (Main) 2024]**

(1) $T^2 \propto R^{5/2}$ (2) $T^2 \propto R^{7/2}$ (3) $T^2 \propto R^{3/2}$ (4) $T^2 \propto R^3$

PGT104

19. A small point of mass m is placed at a distance $2R$ from the centre 'O' of a big uniform solid sphere of mass M and radius R. The gravitational force on 'm' due to M is F_1. A spherical part of radius $R/3$ is removed from the big sphere as shown in the figure and the gravitational force on m due to remaining part of M is found to be F_2. The value of ratio $F_1 : F_2$ is **[JEE (Main) 2025]**



(1) 16 : 9 (2) 11 : 10 (3) 12 : 11 (4) 12 : 9

PGT127

20. A satellite is launched into a circular orbit of radius 'R' around the earth. A second satellite is launched into an orbit of radius 1.03 R. The time period of revolution of the second satellite is larger than the first one approximately by :-

(1) 3 % (2) 4.5 % (3) 9 % (4) 2.5 %

[JEE (Main) 2025]
PGT128

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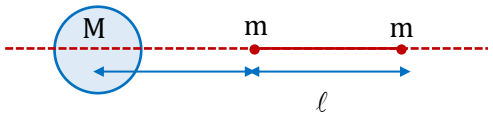
EXERCISE - JEE (Advanced) PYQ

1. A bullet is fired vertically upwards with velocity v from the surface of a spherical planet. When it reaches its maximum height, its acceleration due to the planet's gravity is $1/4^{th}$ of its value at the surface of the planet. If the escape velocity from the planet is $v_{esc} = v\sqrt{N}$, then the value of N is (Ignore energy loss due to atmosphere) **[JEE (Advanced) 2015]**

PGT025

2. A large spherical mass M is fixed at one position and two identical point masses m are kept on a line passing through the centre of M (see figure). The point masses are connected by a rigid massless rod of length ℓ and this assembly is free to move along the line connecting them. All three masses interact only through their mutual gravitational interaction. When the point mass nearer to M is at a distance $r = 3\ell$ from M, the tension in the rod is zero for $m = K\left(\frac{M}{288}\right)$. The value of k is:

[JEE (Advanced) 2015]



PGT026

3. A spherical body of radius R consists of a fluid of constant density and is in equilibrium under its own gravity. If $P(r)$ is the pressure at $r(r < R)$, then the correct option(s) is(are)

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[JEE (Advanced) 2015]

(A) $P(r = 0) = 0$

(B) $\frac{P(r=3R/4)}{P(r=2R/3)} = \frac{63}{80}$

(C) $\frac{P(r=3R/5)}{P(r=2R/5)} = \frac{16}{21}$

(D) $\frac{P(r=R/2)}{P(r=R/3)} = \frac{20}{27}$

PGT027

4. A rocket is launched normal to the surface of the Earth, away from the Sun, along the line joining the sun and the Earth. The Sun is 3×10^5 times heavier than the Earth and is at a distance 2.5×10^4 times larger than the radius of the Earth. The escape velocity from Earth's gravitational field is $v_e = 11.2\ km\ s^{-1}$. The minimum initial velocity (v_s) required for the rocket to be able to leave the Sun-Earth system is closest to (Ignore the rotation and revolution of the Earth and the presence of any other planet) **[JEE (Advanced) 2017]**

(A) $v_s = 22\ km\ s^{-1}$ (B) $v_s = 72\ km\ s^{-1}$ (C) $v_s = 42\ km\ s^{-1}$ (D) $v_s = 62\ km\ s^{-1}$

PGT028

5. A planet of mass M, has two natural satellites with masses m_1 and m_2. The radii of their circular orbits are R_1 and R_2 respectively. Ignore the gravitational force between the satellites. Define v_1, L_1, K_1 and T_1 to be, respectively, the orbital speed, angular momentum, kinetic energy and time period of revolution of satellite 1 ; and v_2, L_2, K_2 and T_2 to be the corresponding quantities of satellite 2. Given $m_1/m_2 = 2$ and $R_1/R_2 = 1/4$, match the ratios in List-I to the numbers in List-II.

[JEE (Advanced) 2018]

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List–I		List–II	
P.	$\frac{\nu_1}{\nu_2}$	1.	$\frac{1}{8}$
Q.	$\frac{L_1}{L_2}$	2.	1
R.	$\frac{K_1}{K_2}$	3.	2
S.	$\frac{T_1}{T_2}$	4.	8

(A) $P \to 4\ ;\ Q \to 2\ ;\ R \to 1\ ;\ S \to 3$ (B) $P \to 3\ ;\ Q \to 2\ ;\ R \to 4\ ;\ S \to 1$

(C) $P \to 2\ ;\ Q \to 3\ ;\ R \to 1\ ;\ S \to 4$ (D) $P \to 2\ ;\ Q \to 3\ ;\ R \to 4\ ;\ S \to 1$

PGT029

6. Consider a spherical gaseous cloud of mass density $\rho(r)$ in free space where r is the radial distance from its center. The gaseous cloud is made of particles of equal mass m moving in circular orbits about the common center with the same kinetic energy K. The force acting on the particles is their mutual gravitational force. If $\rho(r)$ is constant in time, the particle number density $n(r) = \rho(r)/m$ is : [G is universal gravitational constant] **[JEE (Advanced) 2019]**

(A) $\frac{K}{\pi r^2 m^2 G}$ (B) $\frac{K}{6\pi r^2 m^2 G}$ (C) $\frac{3K}{\pi r^2 m^2 G}$ (D) $\frac{K}{2\pi r^2 m^2 G}$

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PGT030

7. The distance between two stars of masses $3M_S$ and $6M_S$ is $9R$. Here R is the mean distance between the centers of the Earth and the Sun, and M_S is the mass of the Sun. The two stars orbit around their common center of mass in circular orbits with period nT, where T is the period of Earth's revolution around the Sun. The value of n is _____. **[JEE (Advanced) 2021]**

PGT031

8. Two spherical stars A and B have densities ρ_A and ρ_B, respectively. A and B have the same radius, and their masses M_A and M_B are related by $M_B = 2M_A$. Due to an interaction process, star A loses some of its mass, so that its radius is halved, while its spherical shape is retained, and its density remains ρ_A. The entire mass lost by A is deposited as a thick spherical shell on B with the density of the shell being ρ_A. If ν_A and ν_B are the escape velocities from A and B after the interaction process, the ratio $\frac{\nu_B}{\nu_A} = \sqrt{\frac{10n}{15^{1/3}}}$. The value of n is _____. **[JEE (Advanced) 2022]**

PGT032

9. Two satellites P and Q are moving in different circular orbits around the Earth (radius R). The heights of P and Q from the Earth surface are h_P and h_Q, respectively, where $h_p = R/3$. The accelerations of P and Q due to Earth's gravity are g_P and g_Q, respectively. If $g_P/g_Q = 36/25$, what is the value of h_Q? **[JEE (Advanced) 2023]**

(A) $3R/5$ (B) $R/6$ (C) $6R/5$ (D) $5R/6$

PGT075

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10. A particle of mass m is under the influence of the gravitational field of a body of mass M ($>> m$). The particle is moving in a circular orbit of radius r_0 with time period T_0 around the mass M. Then, the particle is subjected to an additional central force, corresponding to the potential energy $V_c(r) = m\alpha/r^3$, where α is a positive constant of suitable dimensions and r is the distance from the center of the orbit. If the particle moves in the same circular orbit of radius r_0 in the combined gravitational potential due to M and $V_c(r)$, but with a new time period T_1, then $\left(T_1^2 - T_0^2\right)/T_1^2$ is given by [G is the gravitational constant.]

[JEE (Advanced) 2024]

(A) $\frac{3\alpha}{GMr_0^2}$ (B) $\frac{\alpha}{2GMr_0^2}$ (C) $\frac{\alpha}{GMr_0^2}$ (D) $\frac{2\alpha}{GMr_0^2}$

PGT105

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	D	D	D	B	C	D	C	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	C	B	C	B	D	D	D	A
Que.	21	22	23	24	25	26	27			
Ans.	D	A	C	A	C	A	C			

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9
Ans.	AD	ABC	ACD	C	B	D	B	C	(A-R), (B-Q), (C-Q), (D-P)

EXERCISE (S)

1. 5

2. $\left(\vec{g}=+\frac{\pi G\rho_0 R^3}{6}\left[\frac{1}{(x-(R/2))^2}-\frac{8}{x^2}\right], \vec{g}=-\frac{2\pi G\rho_0 R}{3}\hat{i}\right)$

3. 2

4. 4

5. 7

6. (1.6 hours if it is rotating from west to east, 24/17 hours if it is rotating from east to west)

7. $1\times10^5 J$

8. (a) $T=\frac{2\pi d^{3/2}}{\sqrt{3Gm}}$, (b) 2, (c) 2

9. 2

10. 3

11. 6

12. 2

13. 2

14. 3

15. (a) $\Delta E=\frac{GMm}{2r_1r_2}(r_2-r_1)$ (b) $\Delta E_A=\frac{r_2\Delta E}{r_1+r_2}$, $\Delta E_B=\frac{r_1}{r_1+r_2}\Delta E$

16. 8

17. 8

18. 2

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	2	4	2	4	4	4	2	3	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	3	3	1	1	4	1	3	2

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	7	BC	C	B	D	9	2.30	A	A

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